COMP6009 Assignment Part B

Biogas Plant Intelligent Monitoring and Decision System

Individual Assessment | 20%

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Report and Code Marking Rubric (Total: 20 marks) *5*

Scenario

A Western Australian red-meat processing facility generates organic residues, wastewater, fats and other biodegradable wastes. These materials are treated using an anaerobic digestion system, where microorganisms break down organic matter to produce biogas. The biogas can subsequently be used for energy recovery, while digestate may be recovered as a useful by-product.

Efficient operation is challenging because waste characteristics and loading conditions vary over time. The true biological condition of the digester cannot be observed directly. Operators instead rely on measurements such as methane production, pH, volatile fatty acids and other indicators, which may be noisy or imperfect.

Objective: You are the AI engineer responsible for developing an intelligent monitoring and decision-support system for the facility. Your system must use uncertain sensor information to infer the likely operating condition, track how it changes over time, recommend an appropriate operational response, and evaluate how uncertainty and modelling assumptions affect those recommendations.

System Model

The true operating condition at time *t* is represented by the hidden state:

$$X_t \in \{\text{Stable, Stressed, Critical}\}$$

The monitoring system receives observations:

$$E_t \in \{\text{Normal, Warning, Alarm}\}$$

The intelligent agent can recommend one of three actions:

$$A_t \in \{\text{Continue, Reduce Feed, Intervene}\}$$

A baseline probability model and sensor datasets are supplied. You must implement the required inference and decision-making algorithms yourself.

Task 1 — Probabilistic Modelling and Inference [5 marks]

The true biological condition of the anaerobic digester cannot be directly observed. Instead, the intelligent monitoring system maintains a **belief** about the possible operating states based on uncertain sensor information.

Let the digester state be $X_0 \in$ {Stable, Stressed, Critical}.

Begin with the following **prior belief**:

State	Stable	Stressed	Critical
$P(X_0)$	0.70	0.20	0.10

Sensor 1 provides an indication of digester condition $E_1 \in$ {Normal, Warning, Alarm}, with the following observation model:

True state	Normal	Warning	Alarm
Stable	0.80	0.15	0.05
Stressed	0.25	0.60	0.15
Critical	0.05	0.25	0.70

Sensor 2 monitors the pH condition of the digester $E_2 \in$ {Normal pH, Abnormal pH}, with the following observation model:

True state	Normal pH	Abnormal pH
Stable	0.90	0.10
Stressed	0.50	0.50
Critical	0.15	0.85

Assume that the two sensor observations are **conditionally independent given the true digester state**.

Requirements

- Represent the supplied prior and observation models programmatically and implement Bayesian inference to calculate the posterior belief over the three digester states. Your implementation should be general enough to process different valid observations without hard-coding individual posterior results.
- Investigate single-sensor evidence**.** Determine and compare the posterior belief for each possible observation from Sensor 1. Interpret what these results indicate about the likely condition of the digester.
- Combine evidence from both sensors. Calculate the posterior belief when Sensor 1 reports Warning and Sensor 2 reports Abnormal pH. Compare this posterior with the posterior obtained from Warning alone, and explain the effect of the additional evidence.
- Demonstrate the Bayesian calculation manually. Show the calculation for at least one inference case in your report, including the unnormalised probabilities, normalisation step and final posterior distribution. Verify that your manual result agrees with your Python implementation.
- Interpret the probabilities in the biogas context. Explain the distinction between the true operating state and a sensor observation.
- Critically reflect on the probabilistic model. Discuss limitations and assumptions that could affect inference in a real anaerobic digestion facility. Your discussion should include the conditional-independence assumption between the two sensors and at least one other limitation or modelling assumption.

Task 2 — Monitoring Digester Condition over Time [6 marks]

The digester condition may change over time according to the following baseline transition model:

Current → Next	Stable	Stressed	Critical
Stable	0.80	0.18	0.02
Stressed	0.25	0.60	0.15
Critical	0.05	0.30	0.65

You will be supplied with time-series observations from Sensor 1 introduced in Task 1 for three operating scenarios:

- Scenario A – Stable operation
- Scenario B – Deteriorating operation
- Scenario C – Noisy monitoring

Each scenario contains 30 time steps. You must apply your HMM implementation to all three supplied datasets. **The pH sensor introduced in Task 1 is not used in the HMM analysis.**

Requirements

- Implement Hidden Markov Model (HMM) filtering to calculate $P(X_t \mid e_{1:t})$ at every time step.
  - Explicitly perform the three filtering steps: Prediction -> Observation update -> Normalisation.
  - For the **first observation**, update the supplied prior belief directly. For each subsequent time step, first apply the transition model to predict the next-state belief, then incorporate the new observation.
  - Do not use an HMM library that directly performs the required inference for you.
- For each supplied scenario,
  - calculate the posterior belief distribution over Stable, Stressed, and Critical at every time step;
  - identify the most probable state at every time step;
  - visualise $P(\text{Stable} \mid e_{1:t})$, $P(\text{Stressed} \mid e_{1:t})$ and $P(\text{Critical} \mid e_{1:t})$ over time using clearly labelled plots;
  - Identify and explain **at least two** notable changes in the agent's belief for each scenario. Relate these changes to the observations received and the HMM's previous belief, and compare the behaviour of the model across the three scenarios.
- Briefly discuss at least one assumption or limitation of the HMM model in representing a real anaerobic digestion system, and explain how it may affect the reliability of the inferred state.

Task 3 — Intelligent Operational Decisions [5 marks]

In Task 2, your HMM produced a posterior belief distribution over the digester states at each time step. You will now use these belief distributions to support operational decision-making.

For each time step *t*, let the posterior belief obtained from Task 2 be: $b_t(x) = P(X_t = x \mid e_{1:t})$

Use the following baseline utility model to **determine which operational action has the highest expected utility**. The utilities represent a simplified combination of energy production, throughput, intervention costs and operational risk.

Action	Stable	Stressed	Critical
Continue	+10	+2	−20
Reduce Feed	+5	+6	−5
Intervene	−4	+3	+8

Requirements

- Implement a reusable decision function, e.g. recommend_action(belief, utility_table).
- Using the posterior beliefs generated in Task 2, calculate the expected utility of Continue, Reduce Feed, and Intervene at every time step for all three scenarios.
- Determine the action with the highest expected utility at each time step.
- Visualise or tabulate how the recommended action changes over time for each scenario.
- Identify and explain at least two notable decision points across the scenarios, relating each decision to the HMM belief distribution and the utility model.
- Explain why the action with the highest expected utility may differ from the action associated with the most probable state.
- Briefly discuss at least one assumption or limitation of the utility-based decision model and explain how it could affect the recommended actions in a real anaerobic digestion system.

Deliverables

- Report in **PDF format** (Recommended length: **2,000 words ±10%,** excluding references, captions and appendices.)
  - Recommended Report Structure
    1. **Introduction and Problem Formulation (~150-200 words)**
       Briefly introduce the biogas monitoring problem, the hidden digester states, sensor uncertainty, and the purpose of the intelligent decision-support system. Briefly outline the Bayesian → HMM → expected-utility approach.
    2. **Probabilistic Modelling and Bayesian Inference (~400-450 words)**
       Describe the prior and sensor models and the Bayesian inference method. Include the required manual Bayesian calculation, single-sensor results, combined Sensor 1 + Sensor 2 result, interpretation, and Task 1 assumptions/limitations.
    3. **HMM-Based State Estimation (~550–650 words)**
       Explain the transition model and the Prediction → Observation Update → Normalisation process. Present the results/plots for Scenarios A, B and C, discuss at least two notable belief changes per scenario, compare the three scenarios, and discuss HMM assumptions/limitations.
    4. **Expected-Utility Decision Making (~400-500 words)**
       Explain how the posterior beliefs from Task 2 are used with the utility model. Present recommended actions over time, analyse at least two notable decision points, explain why the maximum-utility action may differ from simply acting on the most probable state, and discuss utility-model assumptions/limitations.
    5. **Conclusion (~150–200 words)**
       Summarise what the intelligent agent was able to infer and recommend. Highlight the main findings across the three scenarios and briefly reflect on the overall usefulness and limitations of the approach.
    6. **References:** Use the IEEE referencing style. References should not count toward the word limit.
    7. **Appendices (optional):** Use these for supplementary material such as detailed numerical results, additional plots, or selected code snippets. The main results and analysis should remain in the report body.
- Python source code containing the Bayesian inference, HMM filtering, expected utility decision-making.
- Results/data files used to produce the reported analyses and visualisations.
- README file with clear instructions for running the code and reproducing the main results.

Report and Code Marking Rubric (Total: 20 marks)

Criteria	Excellent (Full marks)	Satisfactory (Partial marks)	Needs Improvement (Low / No marks)	Weight
1. Probabilistic modelling and Bayesian inference	5.0 - Correct implementation, with accurate calculations, clear interpretation, and insightful discussion of assumptions & limitations.	3.0 – 4.5 - Mostly correct implementation, with reasonable interpretation and some discussion of assumptions & limitations; Minor errors.	0 – 2.5 - Incomplete or incorrect implementation, with limited interpretation and discussion of assumptions & limitations.	/5.0
2. HMM implementation and sequential state estimation	6.0 - Correct implementation; accurate analysis / visualisation of all scenarios with well-justified comparisons; insightful discussion.	3.5 – 5.5 - Mostly correct implementation with appropriate visualisations and interpretation; some comparison and discussion; Minor errors.	0 – 3.0 - Incomplete or incorrect; limited/incorrect results, visualisations or interpretations; little to no meaningful comparison or discussion.	/6.0
3. Expected-utility decision making	5.0 - Correct and reusable implementation; accurate application with clear analysis of recommended actions and notable decision points; strong explanation.	3.0 – 4.5 - Mostly correct implementation, with reasonable analysis and interpretation; some discussion of trade-offs and limitations; Minor errors.	0 – 2.5 - Incomplete or incorrect implementation; limited analysis/justification; weak understanding of probability–utility trade-offs.	/5.0
4. Code quality	2.0 - Code clean, functional, aligned with report	1.0 – 1.5 - Mostly works but not clear	0 – 0.75 - Incomplete or not working	/2.0
5. Report Writing and presentation	2.0 - Clear and well-structured report; proper formatting throughout.	1.0 – 1.5 - Mostly clear; minor issues with writing, structure, or format.	0 – 0.75 Poorly written or presented; hard to follow.	/2.0